

Probability-Based Stock Valuation Framework

v1.0

1. Objective

The objective is to build a repeatable investment framework that values listed equities using probability-weighted scenarios rather than a single-point estimate. The framework combines top-down macro scenario analysis, bottom-up financial statement analysis, Bayesian-style probability updates, peer-group valuation review, and risk-aware portfolio construction.

The final output should answer five questions for each stock:

1. What is the probability-weighted fair value?
2. What is the expected upside/downside distribution?
3. What is the quality and durability of the business?
4. Is the current market price sufficiently attractive relative to the uncertainty?
5. What portfolio weight is justified after considering volatility, correlation, and risk contribution?

2. Core Investment Philosophy

This framework is designed around the following principles:

2.1 Valuation should be probabilistic

A single fair value estimate is usually misleading. For most companies, the real question is not whether the stock is worth exactly X, but how much probability should be assigned to different business outcomes.

The valuation output should therefore be a distribution:

Scenario	Probability	Fair Value	Expected Contribution
Bull case	20%	150	30.0
Base case	45%	110	49.5
Bear case	25%	70	17.5
Stress case	10%	40	4.0
Probability-weighted value	100%	—	101.0

2.2 Macro matters, but company exposure matters more

Macro scenarios should not be applied uniformly to all companies. A higher-rate environment may hurt long-duration growth companies but help banks. A recession may hurt cyclical industrials but leave defensive healthcare or consumer staples relatively resilient.

Therefore, macro analysis should be translated into company-specific operating assumptions.

2.3 Bayesian updating should discipline the investment process

The framework should explicitly track prior beliefs and update them when new evidence arrives.

For example:

Hypothesis	Prior Probability	New Evidence	Evidence Strength	Updated Probability
Temporary earnings trough	35%	Margins stabilizing, inventories normalizing	Moderate positive	42%
Structural decline	25%	Market share still weak	Mild negative	23%
Successful recovery	25%	New product cycle gaining traction	Strong positive	30%
Value trap	15%	Capital allocation remains poor	Mild negative	5%

2.4 Portfolio construction should reflect both expected return and risk contribution

The highest-upside stock is not automatically the largest position. Position size should reflect expected return, confidence, volatility, correlation, liquidity, balance-sheet risk, and downside skew.

3. End-to-End Process

The full process has ten steps:

1. Define macro scenarios and assign probabilities
2. Map each company's exposure to macro variables
3. Analyze company fundamentals from financial reports
4. Build Bayesian hypothesis table for each company
5. Estimate valuation distribution under multiple scenarios
6. Conduct peer-group valuation and quality review
7. Rank stocks using expected return, quality, and uncertainty
8. Select 10-15 best ideas for the portfolio shortlist
9. Construct a risk-aware portfolio
10. Review probabilities and rebalance periodically

4. Macro Scenario Framework

4.1 Default macro scenarios

The macro framework should normally include 4–6 regimes. A practical starting set is:

Macro Scenario	Description	Default Probability
Soft landing / steady growth	Growth slows but remains positive; inflation moderates; rates stable or lower	35%
Reacceleration / reflation	Growth remains strong; inflation sticky; nominal GDP resilient	20%
Mild recession	Earnings decline; unemployment rises; rates fall; credit spreads widen moderately	25%
Stagflation shock	Weak growth with sticky inflation, often commodity-driven	10%
Crisis / liquidity shock	Credit event, geopolitical escalation, or severe funding stress	10%

These probabilities are not fixed. They should be updated at least every one to two weeks.

4.2 Macro variables to monitor

Variable	Why it matters
Policy rates	Discount rate, bank margins, consumer credit, housing
Inflation	Margins, pricing power, real income, central bank reaction
Yield curve	Recession signal, bank profitability, duration valuation
Credit spreads	Risk appetite, refinancing risk, financial stress
PMIs / ISM	Growth momentum and cyclical demand
Employment data	Consumer strength and central bank reaction function
Oil / gas / commodities	Cost pressure, energy profits, inflation impulse
FX	Translation effects, competitiveness, imported inflation
Equity risk premium	Required return and valuation multiples
Earnings revisions	Market-implied fundamental momentum

4.3 Company macro exposure map

Each company should be scored on macro sensitivities:

Exposure Factor	Score Range	Notes
Revenue cyclical	1-5	1 = defensive, 5 = highly cyclical
Operating leverage	1-5	Higher means margins move sharply with revenue
Interest-rate sensitivity	-2 to +2	Positive for banks, negative for long-duration growth / levered firms
Inflation sensitivity	-2 to +2	Positive if pricing power exceeds cost inflation
Commodity sensitivity	-2 to +2	Producer vs consumer of commodities
FX sensitivity	-2 to +2	Benefit or harm from USD/GBP/EUR moves
Credit sensitivity	1-5	Refinancing and spread vulnerability
China / EM sensitivity	1-5	Exposure to China or emerging-market demand
Regulatory sensitivity	1-5	Healthcare, banks, utilities, platforms, defense

5. Bottom-Up Company Analysis

5.1 Business model review

For each company, we should identify:

Area	Questions
Revenue model	What does the company sell? Recurring or transactional? Volume or price driven?
Segment structure	Which segments drive revenue, profit, and growth?
Customer base	Concentrated or diversified? Sticky or price-sensitive?
Cost structure	Fixed vs variable costs? Labor, raw materials, R&D, distribution?
Competitive advantage	Brand, scale, IP, network effects, regulation, switching costs?
Capital intensity	Asset-heavy or asset-light? Maintenance capex burden?
Cash conversion	Does accounting profit convert into free cash flow?
Capital allocation	Buybacks, dividends, M&A, debt repayment, reinvestment?

5.2 Historical financial review

Minimum historical data to review:

Metric	Review Period	Purpose
Revenue growth	5–10 years	Growth durability and cyclicalality
Gross margin	5–10 years	Pricing power and cost pressure
EBIT margin	5–10 years	Operating leverage and efficiency
Net income	5–10 years	Earnings stability
Free cash flow	5–10 years	Cash generation quality
ROIC / ROE	5–10 years	Economic quality
Net debt / EBITDA	5–10 years	Balance-sheet risk
Share count	5–10 years	Buybacks or dilution
Dividend / payout	5–10 years	Capital return discipline

5.3 Quality score

A starting quality score can be based on:

Factor	Weight
ROIC durability	20%
Revenue resilience	15%
Margin stability	15%
Free cash flow conversion	15%
Balance sheet strength	15%
Competitive advantage	10%
Capital allocation	10%

Quality Score = weighted score out of 100.

6. Bayesian Hypothesis Framework

6.1 Investment hypotheses

For each company, define 3–5 mutually understandable hypotheses. They do not need to be perfectly exhaustive, but they should cover the main plausible outcomes.

Common hypothesis types:

Hypothesis Type	Description
Structural compounder	Business can grow earnings and FCF sustainably over time

Hypothesis Type	Description
Temporary slowdown	Current weakness is cyclical or temporary
Successful recovery	Company is in transition but has credible recovery potential
Cyclical value	Earnings are depressed but likely to normalize with the cycle
Mature cash cow	Low growth but strong cash generation and capital returns
Structural decline	Business model is being impaired permanently
Value trap	Cheap valuation is justified by deteriorating fundamentals or poor management
Binary/event-driven	Outcome depends heavily on a specific catalyst or regulatory decision

6.2 Prior probability

Initial prior should be based on:

1. Long-term financial history
2. Industry structure
3. Competitive position
4. Balance sheet
5. Management track record
6. Market valuation relative to history
7. Macro and sector backdrop

Example:

Hypothesis	Prior Probability
Structural compounder	35%
Temporary slowdown	30%
Structural decline	20%
Value trap	15%

6.3 Evidence classification

New evidence should be classified by impact strength:

Evidence Strength	Typical Probability Adjustment
Noise / irrelevant	0%
Weak evidence	±1% to ±3%
Moderate evidence	±3% to ±7%
Strong evidence	±7% to ±15%
Thesis-breaking evidence	>15%

Examples:

Evidence	Likely Strength
One analyst downgrade	Weak
Minor price move without news	Noise / weak
Guidance cut	Moderate to strong
Major customer loss	Strong
Accounting issue	Thesis-breaking
Peer reports severe demand weakness	Moderate
Company gains share in weak market	Strong positive
Balance sheet refinancing risk rises sharply	Strong negative

6.4 Bayesian update table

Every company should have a live table:

Hypothesis	Previous Probability	New Evidence	Evidence Strength	Updated Probability	Change
Structural compounder	40%	Margins resilient despite weak demand	Moderate positive	45%	+5%
Temporary slowdown	30%	Revenue growth still below trend	Mild negative	28%	-2%
Structural decline	20%	No further share loss	Mild negative	17%	-3%
Value trap	10%	Buybacks remain disciplined	Mild negative	10%	0%

7. Valuation Framework

The valuation should combine three anchors:

1. Probability-weighted DCF
2. Earnings-power valuation
3. Peer-adjusted multiple valuation

7.1 Probability-weighted DCF

For each scenario, estimate:

Assumption	Bull	Base	Bear	Stress
Revenue CAGR	—	—	—	—
EBIT margin	—	—	—	—
Tax rate	—	—	—	—
Capex / sales	—	—	—	—
Working capital need	—	—	—	—
Terminal growth	—	—	—	—
WACC / discount rate	—	—	—	—
Equity value per share	—	—	—	—

Then:

Probability-weighted fair value = $\Sigma(\text{probability} \times \text{fair value under each scenario})$

7.2 Earnings-power valuation

This is especially useful for cyclicals, financials, industrials, retailers, energy, and fallen angels.

Core structure:

Normalized revenue
 × normalized EBIT margin
 = normalized EBIT
 - tax
 = normalized NOPAT

Then value using either:

Enterprise value = normalized EBIT × justified EV/EBIT multiple

or:

Enterprise value = normalized NOPAT / WACC

7.3 Peer-group valuation

Peer review should include:

Metric	Company	Peer Median	Premium / Discount	Justification
P/E	—	—	—	—
EV/EBITDA	—	—	—	—
EV/EBIT	—	—	—	—
FCF yield	—	—	—	—
P/B	—	—	—	Mainly for banks/insurers
ROIC	—	—	—	—
Revenue growth	—	—	—	—
EBIT margin	—	—	—	—
Net debt / EBITDA	—	—	—	—

A stock deserves a premium if it has superior growth, ROIC, cash conversion, balance-sheet strength, or lower cyclicality. It deserves a discount if it has weaker durability, higher leverage, poor governance, low cash conversion, or structural decline risk.

7.4 Final valuation blend

Suggested default weighting:

Valuation Method	Weight
Probability-weighted DCF	50%
Earnings-power valuation	25%
Peer multiple valuation	25%

Adjustments:

Company Type	DCF Weight	Earnings Power Weight	Peer Weight
Stable compounder	60%	20%	20%
Cyclical	30%	40%	30%
Bank / insurer	25%	25%	50%
Commodity producer	25%	45%	30%
Early recovery / fallen angel	40%	35%	25%
Mature cash cow	40%	30%	30%

8. Stock Ranking Model

8.0 Market-behavior overlay: sector beta, narrative and retail sentiment

The original framework focused mainly on intrinsic value, business quality, balance sheet and Bayesian fundamental probability. A missing layer is market behavior. Some stocks can move far beyond near-term intrinsic value when a strong sector beta, underdog narrative, retail participation, options activity and institutional FOMO reinforce each other.

This layer should not replace fundamental valuation. It should sit beside it as a separate price-path and timing overlay.

Core principle:

Intrinsic value tells us what the stock may be worth over time.
Market-behavior overlay tells us how far and how fast price may move before fundamentals catch up.

8.0.1 Sector beta / thematic beta

Each stock should receive a sector beta or thematic beta score.

Score	Meaning
1	Low link to an active sector theme
2	Some sector support, but weak narrative
3	Moderate sector beta
4	Strong sector beta
5	Very strong sector beta inside a hot macro/thematic regime

Examples:

Theme	Beneficiaries
AI infrastructure / semiconductors	Semis, memory, foundry, AI servers, power/cooling, data-center supply chain
European rearmament	Defense primes, aerospace, munitions, electronics, shipbuilding
Energy-security / oil shock	Energy majors, oilfield services, LNG, selected utilities
Copper / electrification	Copper miners, grid equipment, power infrastructure
Healthcare policy reset	Managed care, PBMs, hospitals, pharma, depending on policy direction
Payments modernization	Digital wallets, payment processors, stablecoin infrastructure

8.0.2 Narrative strength score

Narrative strength measures whether the stock has a simple, emotionally powerful story that investors can quickly understand.

Score	Meaning
1	No clear narrative
2	Technical or complicated story
3	Understandable but not exciting
4	Strong recovery or regime-change narrative
5	Very strong underdog / national champion / structural transformation narrative

The best narratives usually have:

1. A clear villain or past failure
2. A credible turnaround agent
3. A large future market
4. External validation
5. A simple phrase investors can repeat

8.0.3 Retail sentiment / speculative interest score

Retail sentiment should be tracked using observable proxies:

Proxy	Interpretation
Retail trading flow / brokerage platform ranking	Direct evidence of retail participation
Reddit / Stocktwits / social volume	Narrative heat and crowd attention
Options call volume / call-put skew	Speculative upside chasing
Short interest / days to cover	Potential squeeze fuel
Leveraged ETF flow linked to sector	Sector-level retail amplification
Google Trends / media mentions	Broader public attention
Volume versus average volume	Participation spike
Price gap on news	Reflexive buying pressure

Retail sentiment score:

Score	Meaning
1	Ignored or negative
2	Low interest
3	Neutral / mixed

Score	Meaning
4	Strong retail attention
5	Speculative mania / squeeze-like conditions

8.0.4 Sentiment-adjusted price path

For each stock, maintain two values:

Value	Meaning
Fundamental probability-weighted fair value	Intrinsic value based on scenarios and cash flows
Sentiment-adjusted trading range	Possible near-term price path under strong sector/ narrative/retail momentum

The sentiment-adjusted range should never be treated as intrinsic value. It is a trading and sizing input.

Example:

Stock	Fundamental FV	Sentiment-adjusted range	Interpretation
A	95–115	120–160	Momentum can overshoot fundamentals
B	55–65	45–70	Weak narrative limits re-rating
C	90–100	85–105	Fundamentals dominate, little retail effect

8.0.5 Behavioral overlay score

Add a separate behavioral overlay score out of 20:

Component	Points
Sector / thematic beta	5
Narrative strength	5
Retail sentiment / social attention	5
Options / technical momentum	3
Short squeeze / positioning asymmetry	2
Total	20

This should be reported separately from the 100-point fundamental score.

8.0.6 How to use the overlay

Situation	Action
Fundamental value high, sentiment low	Accumulate patiently
Fundamental value high, sentiment rising	Buy/add more aggressively
Fundamental value fair, sentiment very strong	Momentum trade only; size carefully
Fundamental value low, sentiment euphoric	Avoid chasing; consider trim/sell discipline
Fundamental value improving, sentiment just turning	Best asymmetric setup
Sentiment collapses but thesis intact	Potential add opportunity

This is especially important for fallen angels and regime-change opportunities, where the price path can be reflexive.

8.1 Main scoring categories

Because the preferred style emphasizes fallen angels, deep-value recovery, and cyclical macro opportunities, the screen should reward valuation dislocation and recovery potential more heavily than a pure quality-compounder framework.

Category	Weight
Valuation dislocation / upside	25%
Recovery potential / catalyst path	20%
Business quality / moat durability	15%
Balance sheet / downside protection	15%
Earnings revision potential	10%
Macro scenario skew	10%
Regime-change optionality	5%

Total Score = 100.

For rare major regime-change opportunities, the normal score can be overridden by a separate regime-change score, provided the evidence suggests the opportunity is durable, underappreciated, and not already fully priced.

8.2 Classification

Rating	Description
A	High-conviction compounder at attractive or reasonable price
B	Mispriced recovery / fallen angel with credible path to improvement
C	Cyclical value with favorable risk/reward and macro skew

Rating	Description
D	Cheap but low quality; watchlist only
E	Avoid / value trap / insufficient upside for risk

8.3 Minimum hurdle for portfolio inclusion

A stock should usually require:

Requirement	Suggested Minimum
Expected upside to probability-weighted fair value	>20%
Bear-case downside	Preferably less than 30% unless upside is very asymmetric
Quality score	>60/100, unless special situation
Balance-sheet risk	Manageable
Bayesian thesis probability	Main positive hypothesis >45% or improving strongly
Liquidity	Sufficient for practical portfolio sizing

9. Stock Screening Process

9.0 Preferred opportunity hierarchy

The framework should not treat all types of opportunities equally. The preferred opportunity hierarchy is:

Rank	Opportunity Type	Description
0	Major regime-change opportunity	Rare but highest priority when a long-lasting structural shift creates a multi-year investment opportunity, such as a major technology cycle, defense cycle, energy-security cycle, or regulatory/industrial-policy shift
1	Fallen angels	Previously high-quality or highly regarded companies whose valuation has compressed because of temporary or potentially reversible problems
2	Deep-value recovery	Stocks trading at depressed valuations where earnings, margins, balance sheet, or sentiment may recover
3	Cyclical macro opportunities	Stocks with attractive upside if a macro or sector cycle turns favorably
4	Quality-first value	Good businesses trading at reasonable or modestly discounted valuations

Rank	Opportunity Type	Description
5	Defensive compounders	Durable, lower-risk businesses, useful mainly for portfolio ballast unless valuation is unusually attractive

This means the stock screen should be biased toward asymmetric recovery opportunities, but should still apply quality, balance-sheet, and Bayesian evidence filters to avoid value traps.

9.1 Universe

Initial universe:

1. US large-cap and mid-cap stocks
2. FTSE 100 stocks

Default exclusions:

1. Highly speculative pre-revenue companies
2. Binary biotech without commercial earnings
3. Companies with severe accounting opacity
4. Highly distressed balance sheets unless explicitly treated as special situations
5. Illiquid names

9.2 Initial screen factors

Factor	Desired Direction
FCF yield	High relative to quality and history
P/E or EV/EBIT	Below own history or peer group where justified
ROIC	Stable or improving
Balance sheet	Conservative or improving
Earnings revisions	Stabilizing or improving
Share price drawdown	Large enough to create dislocation, but not thesis-breaking
Business durability	Strong enough to survive adverse scenario
Catalyst visibility	At least some route to value realization

9.3 Candidate buckets

The shortlist should be diversified across investment types, but with clear preference for recovery and dislocation opportunities:

Bucket	Priority	Description
Major regime-change opportunities	Highest, but rare	Long-lasting structural shifts that can create multi-year winners; requires strong evidence that the regime change is real and not already fully priced

Bucket	Priority	Description
Fallen angels	Very high	High-quality or formerly high-quality companies in temporary trouble, with a credible path to recovery
Deep-value recovery	High	Depressed valuations with plausible earnings, margin, balance-sheet, or sentiment recovery
Cyclical macro opportunities	Medium-high	Stocks with depressed earnings or multiples that benefit from a favorable macro or sector turn
Quality-first value	Medium	Good businesses trading at attractive or reasonable valuations
Defensive compounders	Lower, but useful	Lower-upside, durable cash-flow names used mainly for diversification, downside protection, or when unusually mispriced

10. Periodic Review and Bayesian Updating

10.1 Review hierarchy

Review Type	Frequency	Purpose
Light probability update	Weekly or bi-weekly	Incorporate macro and news flow
Portfolio rebalance review	Monthly	Adjust weights if required
Full model refresh	Quarterly	Rebuild models after earnings
Event-driven review	As needed	Major news, guidance, M&A, regulatory events

10.2 Weekly / bi-weekly review

Focus areas:

1. Macro data and central bank communication
2. Rates, inflation, credit spreads, commodities and FX
3. Sector news
4. Company-specific news flow
5. Earnings revisions and analyst estimate changes
6. Share price and valuation moves
7. Options-implied volatility and short interest if useful
8. Any change in investment thesis

Output table:

Stock	Previous View	New Evidence	Probability Change	Valuation Change	Action
A	Buy	Margins better than expected	Bull case +5%	Fair value +4%	Add

Stock	Previous View	New Evidence	Probability Change	Valuation Change	Action
B	Hold	Sector data weaker	Bear case +7%	Fair value -6%	Trim
C	Buy	No major news	0%	0%	Hold
D	Watchlist	Price down, thesis unchanged	Upside improved	0%	Consider entry

10.3 Probability freshness rule

If a stock has not been reviewed recently, confidence should decay.

Time Since Last Review	Action
Less than 2 weeks	No decay
2–4 weeks	Small confidence haircut
1–2 months	Medium confidence haircut
More than 2 months	Full refresh required before adding capital

10.4 Evidence update discipline

Avoid overreacting to noise. Every probability update should identify:

1. What changed?
2. Is it new information or already priced in?
3. Which hypothesis does it support?
4. How strong is the evidence?
5. Does it change valuation, risk, or only sentiment?
6. Does it require a portfolio action?

11. Portfolio Construction

11.1 Target portfolio

Default portfolio structure:

Parameter	Suggested Default
Number of stocks	10–15
Single stock max weight	10–12%
Single stock min weight	3%
High-volatility stock max weight	5–7%
Sector max weight	25–30%

Parameter	Suggested Default
Deep-value / turnaround bucket max	20–25%
Defensive / compounder bucket target	30–50%

11.2 Position sizing formula

A practical starting formula:

$$\text{Target weight} \propto \text{Conviction Score} \times \text{Expected Return} / \text{Volatility Contribution}$$

Where:

Input	Meaning
Conviction score	Confidence in thesis and quality of evidence
Expected return	Probability-weighted upside including dividends
Volatility contribution	Volatility adjusted for correlation with portfolio
Downside penalty	Haircut for large bear-case losses
Liquidity penalty	Haircut for weaker liquidity

11.3 Risk contribution review

For each holding:

Stock	Weight	Volatility	Correlation	Expected Return	Risk Contribution	Action
A	8%	25%	0.60	30%	7%	Hold
B	5%	45%	0.75	45%	10%	Cap weight
C	10%	18%	0.50	20%	6%	Hold

11.4 Rebalance triggers

Trigger	Possible Action
Expected upside falls below required return	Trim or exit
Probability of negative thesis rises materially	Reduce weight
Positive thesis probability falls by more than 10 percentage points	Deep review
Stock reaches fair value	Trim
Better risk-adjusted opportunity appears	Switch capital
Risk contribution becomes excessive	Reduce weight

Trigger	Possible Action
Sector concentration breaches limit	Rebalance
Volatility rises without valuation compensation	Reduce or hedge
Price falls but thesis improves	Add selectively
Price falls and thesis weakens	Do not average down automatically

12. Company Report Template

Each deep-dive report should use the following structure:

1. Investment summary
2. Business description
3. Segment-level financial analysis
4. Historical revenue, margin, ROIC and FCF review
5. Balance sheet and capital allocation
6. Competitive advantage / moat
7. Key uncertainties
8. Bayesian hypothesis table
9. Macro scenario sensitivity
10. Valuation distribution
11. Peer-group comparison
12. Expected return and downside risk
13. Final rating
14. Portfolio weight recommendation

12.1 Investment summary template

Item	Output
Current price	—
Probability-weighted fair value	—
Expected upside	—
Bear-case downside	—
Dividend yield	—
Total expected return	—
Quality score	—
Bayesian positive-thesis probability	—
Final rating	—

Item	Output
Suggested portfolio weight	—

12.2 Valuation summary template

Scenario	Probability	Fair Value	Implied Upside / Downside	Key Assumption
Bull	—	—	—	—
Base	—	—	—	—
Bear	—	—	—	—
Stress	—	—	—	—
Probability-weighted	100%	—	—	—

12.3 Peer review template

Company	P/ E	EV/ EBITDA	FCF Yield	ROIC	Net Debt / EBITDA	Growth	Comment
Target company	—	—	—	—	—	—	—
Peer 1	—	—	—	—	—	—	—
Peer 2	—	—	—	—	—	—	—
Peer 3	—	—	—	—	—	—	—
Peer median	—	—	—	—	—	—	—

13. Decision Rules

13.1 Buy

A stock can be considered for purchase if:

1. Expected upside is attractive relative to downside.
2. Positive thesis probability is high or rising.
3. Business quality is acceptable or better.
4. Balance sheet is not a major threat.
5. Peer review confirms valuation is not obviously misleading.
6. Portfolio risk contribution is acceptable.

13.2 Add

Add to an existing position if:

1. Price falls but thesis is unchanged or stronger.
2. Probability-weighted upside increases.

3. Risk contribution remains within limit.
4. No superior opportunity has emerged.

13.3 Hold

Hold if:

1. Thesis remains intact.
2. Upside remains acceptable.
3. Risk contribution is stable.
4. No major probability update is required.

13.4 Trim

Trim if:

1. Stock approaches fair value.
2. Expected return falls below hurdle.
3. Position becomes too large due to price appreciation.
4. Risk contribution becomes excessive.
5. Macro scenario exposure becomes too concentrated.

13.5 Exit

Exit if:

1. Original thesis is broken.
2. Positive hypothesis probability falls materially.
3. Accounting or governance risk rises sharply.
4. Balance-sheet risk becomes unacceptable.
5. Better alternatives offer superior risk-adjusted return.

14. Initial Implementation Plan

Phase 1: Finalize framework

Deliverables:

1. Macro scenario template
2. Company report template
3. Bayesian probability update template
4. Valuation model assumptions template
5. Portfolio construction template

Phase 2: Run first universe scan

Universe:

1. US large and mid-cap stocks
2. FTSE 100 stocks

Output:

1. Top 10–15 candidate list
2. Categorization by investment type
3. Initial valuation dislocation estimate
4. Initial quality and risk score
5. Priority order for deep-dive reports

Phase 3: Produce detailed company reports

For each candidate:

1. Full business analysis
2. Financial report review
3. Bayesian hypothesis table
4. Scenario valuation
5. Peer review
6. Portfolio suitability assessment

Phase 4: Build portfolio

Output:

1. Target weights
2. Expected return contribution
3. Volatility and correlation review
4. Risk contribution table
5. Scenario stress test
6. Rebalance rules

Phase 5: Maintain live process

Output every one to two weeks:

1. Macro probability update
2. Company probability updates
3. Watchlist changes
4. Portfolio action recommendations

Monthly:

1. Rebalance review
2. Updated risk contribution
3. Updated target weights

Quarterly:

1. Full model refresh after earnings
2. Updated valuation distributions
3. Updated portfolio ranking

15. Screening Scorecard v1.0

This section converts the investment preference into a practical stock-screening model. The purpose of the screen is not to make final investment decisions. The purpose is to rank the investable universe and identify the 25–40 names most worth deeper research.

15.1 Overall score

Each stock receives a score out of 100:

Category	Weight
Valuation dislocation / upside	25%
Recovery potential / catalyst path	20%
Business quality / moat durability	15%
Balance sheet / downside protection	15%
Earnings revision potential	10%
Macro scenario skew	10%
Regime-change optionality	5%
Total	100%

The score is designed to favor fallen angels, deep-value recovery, and cyclical macro opportunities. Defensive compounders can still pass the screen, but only if valuation is unusually attractive or portfolio diversification benefit is high.

15.2 Valuation dislocation / upside score: 25 points

This measures whether the stock is meaningfully cheap relative to its normalized earnings power, peer group, and own history.

Sub-factor	Points	What to look for
Discount to historical valuation range	6	P/E, EV/EBIT, EV/EBITDA or P/B materially below 5–10 year history
Discount to peer group	5	Valuation below comparable peers after adjusting for quality and growth
Free cash flow yield	5	High FCF yield relative to market, sector, and own history
Normalized earnings upside	5	Current earnings depressed versus plausible mid-cycle earnings power
Dividend / buyback support	4	Capital return provides downside support or accretive repurchase optionality

Sub-factor	Points	What to look for
Total	25	—

High score example: a former market favorite trading at a large discount to history because margins are temporarily depressed.

Low score example: a cheap-looking stock where earnings are structurally declining and cash flow is poor.

15.3 Recovery potential / catalyst path score: 20 points

This is the core score for fallen angels and deep-value recovery names. It asks whether there is a credible reason for the market to re-rate the stock.

Sub-factor	Points	What to look for
Margin recovery potential	5	Cost cuts, mix improvement, input cost relief, pricing recovery
Revenue stabilization	4	End-market demand bottoming, order book improving, churn stabilizing
Product / pipeline / innovation catalyst	4	New product cycle, drug pipeline, technology transition, platform improvement
Management action	3	Restructuring, asset sales, buybacks, leverage reduction, strategic reset
Sentiment reset potential	2	Expectations sufficiently low that even stabilization can help
Timing visibility	2	Catalyst likely within 6–24 months rather than vague long-term hope
Total	20	—

High score example: a fallen angel with depressed margins, credible restructuring, and improving order trends.

Low score example: a cheap stock with no visible catalyst and continued estimate downgrades.

15.4 Business quality / moat durability score: 15 points

This prevents the recovery screen from becoming a value-trap screen.

Sub-factor	Points	What to look for
ROIC / ROE quality	4	Returns above cost of capital over a cycle

Sub-factor	Points	What to look for
Competitive position	3	Brand, scale, IP, cost advantage, distribution, switching costs
Margin structure	3	Resilient margins or clear reason margins can recover
Revenue durability	3	Recurring demand, installed base, regulation, mission-critical products
Cash conversion	2	Earnings convert into free cash flow
Total	15	—

The framework does not require every stock to be a high-quality compounder. But even recovery stocks need enough quality to make recovery plausible.

15.5 Balance sheet / downside protection score: 15 points

This is especially important for deep-value and cyclical stocks. A cheap stock with excessive leverage can become an equity wipeout.

Sub-factor	Points	What to look for
Net leverage	4	Net debt / EBITDA manageable versus sector norms
Interest coverage	3	EBIT or EBITDA comfortably covers interest expense
Debt maturity profile	3	No near-term refinancing wall under stressed conditions
Liquidity / cash buffer	2	Sufficient cash and credit lines
Asset backing / liquidation value	2	Tangible assets, working capital, or hidden assets support downside
Pension / off-balance-sheet risk	1	No major hidden liabilities
Total	15	—

For banks and insurers, replace leverage metrics with capital ratios, asset quality, reserving strength, liquidity, and regulatory capital buffer.

15.6 Earnings revision potential score: 10 points

This helps avoid buying too early into a falling earnings cycle.

Sub-factor	Points	What to look for
Estimate stabilization	3	Consensus downgrades slowing or stopping
Guidance credibility	2	Management guidance realistic and not repeatedly cut

Sub-factor	Points	What to look for
Peer read-through	2	Competitors showing improving or stable trends
Operating momentum	2	Volumes, pricing, bookings, or backlog improving
Surprise potential	1	Low expectations create scope for positive surprise
Total	10	—

A stock can still qualify if revisions are negative, but only if valuation already discounts a severe outcome and the Bayesian recovery probability is improving.

15.7 Macro scenario skew score: 10 points

This measures whether the stock benefits from or survives the macro scenarios we assign the highest probabilities to.

Sub-factor	Points	What to look for
Resilience in base case	3	Acceptable return if macro remains ordinary
Upside in favorable macro case	3	Clear earnings or multiple upside if cycle turns
Downside survival in adverse case	2	Balance sheet and cash flow survive recession/stress
Diversification benefit	2	Adds differentiated macro exposure to the portfolio
Total	10	—

For cyclical macro opportunities, this score can be high even if business quality is only average, provided downside survival is credible.

15.8 Regime-change optionality score: 5 points

This captures rare but potentially very powerful multi-year opportunities.

Sub-factor	Points	What to look for
Structural durability	2	Change likely lasts several years, not just one quarter
Underappreciated earnings impact	1	Market has not fully priced future earnings power
Policy / capex / technology support	1	Supported by industrial policy, technology adoption, or capex cycle
Breadth of opportunity	1	Benefits multiple companies or an entire value chain
Total	5	—

Examples may include major semiconductor cycles, defense rearmament, energy security, electrification, AI infrastructure, reshoring, or regulatory market restructuring.

16. Penalty Framework

A stock can have an attractive headline score but still be penalized for specific risks.

16.1 Standard penalties

Risk	Suggested Penalty
Accounting quality concern	-5 to -20
Governance concern	-5 to -15
Excessive leverage	-5 to -20
Large near-term refinancing risk	-5 to -20
Structural revenue decline	-5 to -15
Value trap characteristics	-5 to -20
Highly binary outcome	-5 to -15
Poor liquidity	-3 to -10
Severe regulatory risk	-5 to -20
Extreme valuation despite good story	-5 to -15

16.2 Automatic deep-review flags

The following do not automatically exclude a stock, but they require careful review before inclusion:

1. Negative free cash flow for more than two consecutive years
2. Net debt / EBITDA above sector comfort range
3. Repeated guidance cuts
4. Auditor change, restatement, or accounting controversy
5. Major management turnover
6. Customer concentration above acceptable level
7. Business model disruption from new technology
8. Large pension or litigation liability
9. Political, sanctions, or regulatory overhang
10. Falling share price with worsening fundamentals rather than improving risk/reward

17. Screening Output Format

The first screen should produce a 25–40 stock candidate list with the following fields:

Field	Description
Ticker	Stock ticker

Field	Description
Company	Company name
Market	US / UK
Sector	GICS or similar sector
Opportunity bucket	Fallen angel, deep-value recovery, cyclical macro, quality value, defensive, regime-change
Total score	Score out of 100 after penalties
Valuation score	Out of 25
Recovery score	Out of 20
Quality score	Out of 15
Balance sheet score	Out of 15
Earnings revision score	Out of 10
Macro score	Out of 10
Regime-change score	Out of 5
Key thesis	One-sentence investment case
Main risk	One-sentence downside risk
Deep-dive priority	High / medium / low

18. Top-Down to Bottom-Up Screening Workflow

The practical workflow should be:

1. Update macro scenario probabilities
2. Identify favored sectors and macro exposures
3. Run broad valuation and quality screen
4. Identify large drawdowns and valuation dislocations
5. Separate fallen angels from value traps
6. Add regime-change watchlist names
7. Score all candidates using the scorecard
8. Select 25-40 preliminary candidates
9. Narrow to 10-15 deep-dive candidates
10. Produce full probability-based reports

19. Immediate Next Step

The next practical step is to run the first broad scan of US large/mid-cap stocks and FTSE 100 stocks using the scorecard above.

The expected output should be:

1. A preliminary 25–40 name candidate list
2. Classification by opportunity type
3. Initial scoring using the v1.0 scorecard
4. A narrower recommended 10–15 name deep-dive list
5. Priority order for detailed company reports